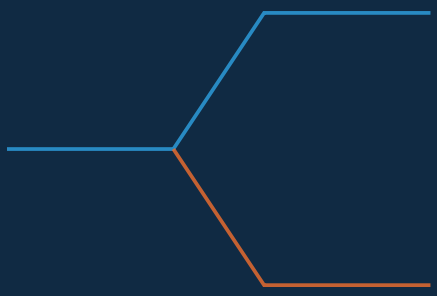
Epoxy / Amine Compound Funnel

Non-ranking digital proposals to CRO feasibility-review packets



Detailed design, module-by-module status, evidence gaps, acceptance record, throughput, and 20 final observed compound audit cards. LOCKED_INTERNAL / NO RANKING / NO LAB EXECUTION

01 / EXECUTIVE DECISION

The funnel is implemented; promotion is not yet earned

8 / 21	210	13 / 29	0
Phase 0 / pre-Phase-0 today	workspace tests passing	locked diagnostics passing	scientific ranks emitted

Decision in one sentence

The software boundaries, locked manifests, typed property-to-formulation chain, signed non-executable CRO packet, sparse Phase-0 runtime grants, ORD acquisition verifier, rights workflow, and non-ranking campaign CLI now exist and pass the ordinary workspace suite; the four module movements are correctly withheld because ORD objects, the provenance-built RDKit runtime, rights receipts, independent signatures, locked campaign evidence, and the 96-attack/32-control review have not been supplied.

What is genuinely good

- Fail-closed authority is intact: every observed envelope has `usable_by_optimizer:false`; no objective or rank is emitted.
- Exact typed handoffs prevent candidate injection of molecular weight, EEW, AHEW, group counts, or authoritative density.
- The v4 catalog is parent-bound to immutable v3 and declares all 550 ORD LFS objects and 1,181,061,030 bytes.
- Twenty audit probes traverse 12 controls twice with identical result hashes.
- Existing data and global audits pass; v3 retains 738,149 normalized rows and 42,158 explicit rejections.

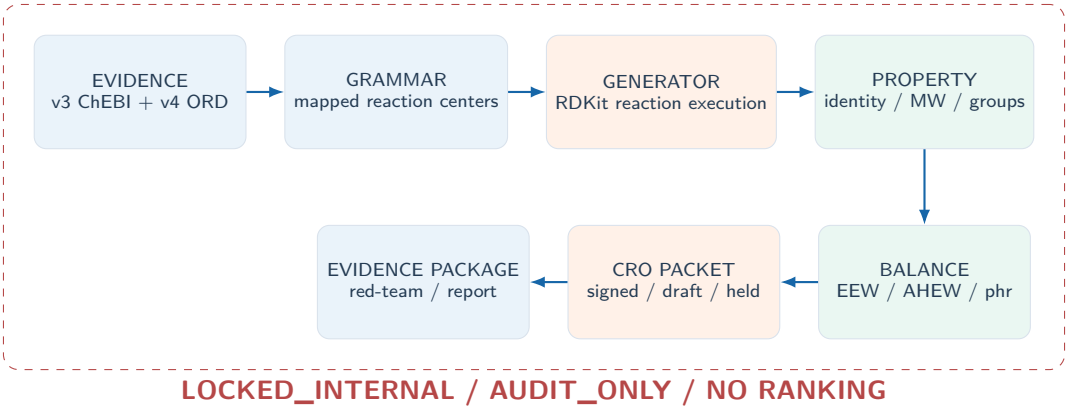
What is not good enough

- v4 has one acquisition unit pending and zero acquired ORD objects; a catalog of pointers is not the reaction corpus.
- The exact C++ RDKit build manifest and library hashes are not installed; feature tests fail closed at the build boundary.
- Rights verification has no active non-development reviewer identity.
- No 24-structure campaign cohort, 12 verified pair packets, or campaign KPI denominator exists yet.
- The requested sealed independent attacks and 32 valid controls have not been received or executed.

Truth boundary. The 20 structures at the end are pre-campaign audit probes, not ORD-generated campaign outputs. “Final” means final observed card at this implementation snapshot. It does not mean synthesized, safe, available, novel, qualified, or selected.

02 / OVERALL DESIGN

A hash-bound funnel with authority separated from computation



Scientific data plane

1. Acquire and verify complete snapshots; normalize without erasing rejects.
2. Split reaction, canonical-family, and scaffold components before extracting grammar.
3. Execute only host-authorized reaction SMARTS on the exact backend.
4. Recompute identity and molecular facts from the resulting graph.
5. Select cohorts by canonical hash within role/scaffold strata—never by predicted performance.
6. Compute theoretical equivalent balance and bind draft CRO packets to candidate and formula hashes.

Authority plane

1. Rights, evaluation, and review use three distinct authorized signers.
2. `EngineEvidenceContextV3` activates sparse module-specific grants; unrelated novelty or Process-QC grants are not prerequisites.
3. Each grant binds movement, evaluation, review, rights overlay, benchmark, model/grammar, inputs, and claim locks.
4. Every CRO packet is `DRAFT_FOR_CRO_FEASIBILITY_REVIEW`, signed, non-executable, and held for material, chemist/EHS, and CRO acceptance.
5. Movement is atomic and module-specific. Failed science remains visible and cannot be converted into authority.

The central design choice
Computation and authority are deliberately orthogonal. A module can calculate a diagnostic value while remaining unable to rank, recommend procurement, authorize synthesis, claim safety, or release a formulation. Conversely, a signature cannot replace missing chemistry, data, denominators, or negative results.

Boundary	Accepted	Rejected / quarantined
Grammar	Signed host artifact bound to ORD IDs, family, SMARTS, constraints, and hashes	Candidate-supplied SMARTS, source IDs, grammar, or substituted artifact
Molecule facts	Exact host RDKit graph result and its hash	Candidate MW, formula, EEW/AHEW, group count, identity override
Protocol authority	Locked protocol reference for numeric thresholds	Packet-authored performance threshold or implied execution

03 / EVIDENCE AND TRUST

Immutable catalogs are ready; the external evidence is not

Layer	Audit state	Measured content	Interpretation
feedback/v3	PASS	6/6 units; 738,149 rows; 42,158 rejects	Existing corpus is immutable. ChEBI release 252 remains available, but property rights receipt is absent.
feedback/v4	PASS catalog / PENDING data	1 unit declared; 0 acquired; 550 ORD objects; 1.181 GB	Parent hash binds v3. Every official LFS OID, byte size, path, and expected raw SHA is declared.
RDKit C++	ABSENT	Release 2026.03.2, commit `42fe8c...`, source SHA pinned	Reproducible container recipe exists; build and library receipts do not. Feature compilation refuses missing paths.
Rights identities	ABSENT	request / offline-sign / import / verify flow implemented	Trust store has no active independent rights reviewer; no repository private key is accepted.

ORD_GIT_LFS_TREE hard gates

- exact official repository and Git commit `ddb0d257...`;
- exact 550 `.pb.gz` path set—no unexpected or missing object;
- pointer OID equals retrieved raw SHA-256 and declared size;
- no symlink, unresolved pointer, partial retrieval, corrupt object, or companion-file substitution;
- immutable content-addressed installation plus signed acquisition receipt;
- re-audit of the complete installed set before normalization.

Leakage-safe grammar policy

- Parse only explicit, single-component, atom-mapped reactants and products with provenance.
- Derive a well-defined reaction center or reject with a reason.
- Partition connected reaction identity, canonical compound family, and scaffold components before extraction.
- Keep ORD-derived grammar in the CC-BY-SA-4.0 evidence layer; do not copy it into MIT source.
- Export attribution and share-alike metadata with every derived grammar and CRO evidence bundle.

Observed failure is correct

`compound campaign run -locked ...` stops at "complete ORD Git LFS snapshot has not been acquired." The RDKit feature build stops at missing `RDKIT_INCLUDE_DIR`. Both are desired fail-closed behavior; neither is campaign success.

Trust ceremony still required

Day zero must register and prove possession of three distinct, non-revoked, non-development Ed25519 identities: rights reviewer, evaluator, and independent reviewer. Cross-role reuse, stale receipts, same-signer review, wrong snapshot, wrong module, stripped exclusions, or repository-local private material is rejected.

Licensing posture. ORD snapshot and derived grammar are treated as CC-BY-SA-4.0 evidence. ChEBI release 252 is treated as CC-BY-4.0 input. This is an implementation control, not a legal opinion; the signed rights reviewer remains the authority required by the movement gate.

04 / PORTFOLIO REGISTER

Every native module and its current phase

Measured by `portfolio phase0 status -all`. Eight modules have signed Phase-0 movements; 21 remain pre-Phase-0. All 29 are confined to non-ranking authority.

Module	P	Primary signal
barrier-corrosion-score	PRE	S/R/E
candidate-contract	P0	complete
cro-protocol-packager	PRE	E
cure-potlife-gate	PRE	R/E
design-rule-score	PRE	S/R/E
envelope-gates	PRE	E
evidence-intake-validator	P0	complete
formulation-arithmetic	PRE	E
formulation-universe	PRE	E
ingredient-library	PRE	S/R/E
ip-fto-guard	PRE	R/E
known-answer-benchmarks	PRE	S/R/E
model-card-authority	P0	complete
molecule-design-gate	P0	complete
molecule-generator	PRE	ORD/R/E
molecule-novelty	PRE	S/R/D/E
molecule-property	PRE	S/R/E
molecule-role-ranker	PRE	R/E
optimizer-orchestrator	PRE	E
process-qc-twin	PRE	R auth
public-corrosion-benchmark	PRE	S/R/E
redteam-audit	P0	complete
report-packager	P0	complete
run-provenance	P0	complete
solvency-score	PRE	S/R/E
source-ledger	P0	complete
sourceability-ehs	PRE	R/E
tolerance-robustness	PRE	S/R/E
viscosity-score	PRE	S/R/E
Portfolio: 8 Phase 0 / 21 pre-Phase-0		

Signal key: S source lock; R rights; D required data; E external evaluation; auth cryptographic authorization.

Promotion target	Phase	Locked diagnostic	Status interpretation
molecule-property	PRE	FAIL / AUDIT	Implementation present; exact backend, locked ontology, rights overlay, 300-structure benchmark, evaluator and reviewer absent.
molecule-generator	PRE	FAIL / ROADMAP	Authorized reaction execution present; ORD objects, extracted grammar, 24-structure cohort, rights and review absent.
formulation-arithmetic	PRE	PASS / COMPLETE	Deterministic legacy diagnostic passes; typed host-facts balance is implemented, but external evaluation and movement are absent.
cro-protocol-packager	PRE	PASS / COMPLETE	Deterministic packet diagnostic passes; V2 signed draft/holds implemented, but candidate-bound 12-packet campaign and movement are absent.

Why 12 / 17 was not written into the ledger

The plan explicitly forbids substituting synthetic records or weakening gates. A passing implementation test is not an external evaluation signature; a report-only structure is not an ORD-generated proposal; and a deterministic CRO fixture is not one of the required 12 candidate-bound feasibility packets. The correct portfolio therefore remains 8 / 21.

05 / MODULE DEEP DIVE I

Molecule property and molecule generator

01 / AMPHORA-MOLECULE-PROPERTY

Graph facts, not guessed descriptors

Current status: PRE **Diagnostic:** FAIL / AUDIT

Implemented

- Exact-backend boundary for canonical isomeric identity, formula, molecular weight, fragments, stereochemistry, oxirane count, reactive amine-H count, and scaffold.
- Backend version and ontology version are carried in `MoleculeFactsV1`; mismatches fail typed downstream arithmetic.
- EEW and AHEW derive only as

$$\text{EEW} = M/n_{oxirane},$$
$$\text{AHEW} = M/n_{amine-H}.$$

  Zero groups abstain; nonfinite, negative, wildcard, polymer, invalid-valence, and multicomponent cases are rejected or quarantined.

GOOD Exact graph-derived facts are structurally separated from candidate claims. Unit and hash provenance are retained.

BAD / BLOCKED No provenance-built C++ runtime receipt is installed. The required 300-structure locked benchmark and ChEBI rights receipt do not exist.

LEFT Build the pinned runtime; sign ChEBI rights; run identity/stereo/fragment/invalid/epoxy/amine controls; report formula agreement, mass error distribution, confusion matrix, abstentions, numerator, denominator, rejects, and negative results; obtain evaluator and reviewer signatures.

02 / AMPHORA-MOLECULE-GENERATOR

Authorized reactions with exact lineage

Current status: PRE **Diagnostic:** FAIL / ROADMAP

Implemented

- Reaction grammar now carries family, reaction SMARTS, ORD source reaction/dataset IDs, applicability constraints, and a stable artifact hash.
- Candidate JSON cannot provide or replace grammar. `EngineEvidenceContextV3` grants generator access independently of novelty and Process-QC.
- The RDKit C++ bridge parses and runs chemical reactions, sanitizes outputs, rejects multicomponent products, sorts unique products, and binds parent, seed, grammar, template, source IDs, role, facts, and scaffold.
- Without a verified runtime grant, behavior remains `QUARANTINED_NO_STRUCTURE_CHANGE`.

GOOD The former ID-only mutation path has an authorized structure-changing replacement, with deterministic duplicate and applicability accounting.

BAD / BLOCKED There is no acquired ORD object set, parsed reaction corpus, extracted held-out grammar, runtime grant, or generated cohort. The 24/8/8/6 thresholds are therefore unmet.

LEFT Acquire all 550 objects; parse and record exclusions; build leakage-safe components; extract/share-alike grammar; generate at most 96 attempts; retain every duplicate and applicability rejection; meet thresholds without synthetic substitutions; obtain independent replay.

Important distinction. Python RDKit 2026.03.2 was used only to draw the structures in this report and to reject unparseable illustration input. It did not create host-authoritative campaign facts. The campaign requires the separately pinned C++ build, commit, flags, compiler/dependency manifest, and library hashes.

06 / MODULE DEEP DIVE II

Equivalent balance and CRO feasibility packaging

03 / AMPHORA-FORMULATION-ARITHMETIC

Typed theoretical stoichiometry

Current status: PRE **Locked diagnostic:** PASS / COMPLETE

Implemented contract: `EquivalentBalanceV1`

- Accepts only two host-derived `MoleculeFactsV1` values and their stable hashes.
- Requires exact backend and ontology, an epoxide-bearing resin, and reactive-amine-H hardener.
- Computes resin/hardener mass, EEW, AHEW, epoxide equivalents, amine-H equivalents, hardener phr, equivalent ratio, total mass, and unit provenance.
- Persists `candidate_property_override_accepted:false`, `ranking_eligible:false`, and `AUDIT_ONLY`.

For resin mass m_r and target ratio r:

$$n_e = m_r/\mathrm{EEW}, \qquad m_h = n_e\,\mathrm{AHEW}\,r,$$
$$\mathrm{phr} = 100 m_h/m_r.$$

GOOD Candidate MW, EEW, AHEW, and group-count injection has no typed entry point. Revalidation recomputes the full record.

BAD / BLOCKED A deterministic fixture pass is not 12 real pair calculations. No campaign facts exist, so no authoritative EEW/AHEW pair can be emitted.

LEFT Run 12 deterministic hash-selected pairs after the generated cohort passes; retain all zero-group, unit, hash, nonfinite, and inversion failures; sign evaluation and replay.

04 / AMPHORA-CRO-PROTOCOL-PACKAGER

A signed draft that cannot execute

Current status: PRE **Locked diagnostic:** PASS / COMPLETE

Implemented contract: `CroProtocolPackageV2`

- Binds candidate JSON/hash, formula hash, and exact `EquivalentBalanceV1`.
- Requires blinding, custody chain, controls, required outputs, unresolved material/safety fields, chemist/EHS hold, and CRO-acceptance hold.
- Status is exactly `DRAFT_FOR_CRO_FEASIBILITY_REVIEW`; `execution_authorized:false`.
- Eight mandatory locks prevent synthesis, safety, performance, qualification, procurement, release, ranking, and execution claims.
- Numeric power/decision thresholds appear only through a separately locked protocol reference.
- Packet ID and Ed25519 signature are verified; any mutation invalidates the package.

GOOD The packet is candidate-bound, formula-bound, signable, byte-stable, and explicitly non-executable.

BAD / BLOCKED The 12 required campaign packets do not exist. The current locked diagnostic uses controlled fixtures and cannot imply CRO acceptance or results.

LEFT Create 12 packets only after typed balances exist; verify clean byte replay, signer identity, custody/controls/holds, and prohibited-claim scan; package CC-BY-SA attribution; obtain evaluation and independent review.

No laboratory hand-wave

A feasibility packet is a structured question for a CRO, not an instruction to execute and not evidence of feasibility. Material availability, chemical review, EHS review, CRO acceptance, detailed locked protocol, synthesis, and testing all remain outside this milestone.

07 / COMPLETION PATH

What remains, red-team acceptance, and measured rate

MEASURED CONTROL-PLANE RATE

3.03 compounds / second

Single worker; 20 composite cards; 60 lane evaluations; 240 module envelopes; 6.596 s first pass; second pass verified identical result hashes.

How to read that rate. It is the measured throughput of current control-plane traversal with chemistry facts unavailable. It is *not* reaction generation rate, exact RDKit property rate, ORD parsing rate, formulation optimization rate, or CRO packet throughput. The pinned C++ campaign benchmark must report those stage rates separately, with warm/cold conditions, worker count, hardware, rejects, and denominators.

Locked completion sequence

1. Install complete ORD objects and signed rights overlay; build exact RDKit C++ and its manifest.
2. Benchmark property on ≥ 300 structures and extract leakage-safe reaction grammar.
3. Generate ≥ 24 unique valid structures: ≥ 8 epoxide, ≥ 8 reactive amine, ≥ 6 scaffold-pair groups.
4. Produce 12 typed balances and 12 signed draft CRO packets.
5. Execute 64 internal malicious cases, reveal/execute 32 sealed independent cases, and accept all 32 valid controls.
6. Obtain evaluator and independent replay signatures; promote property, generator, arithmetic, and CRO packager one movement at a time with a global audit between movements.

Exact red-team gate

Required outcome	Now
96 malicious cases executed	0 campaign
32 valid controls accepted	0 campaign
Silent malicious acceptance	not measured
Unauthorized rank/objective	0 observed
Stripped claim lock	0 observed
Byte-identical clean replay	20 probes pass

The already promoted red-team control has a much larger 30,000-negative deterministic corpus with independent reimplementation. That is valuable baseline coverage, but it does not substitute for the chemistry-specific 96/32 campaign, including sealed reviewer payloads.

Acceptance attempt ledger

Command family	Observed
format / workspace test	PASS, 210
global native audit	PASS, 29
data v3 / v4 audit	PASS / PASS
locked diagnostics	13 pass / 16 fail
pinned RDKit feature suite	FAIL-CLOSED
rights property / generator	FAIL-CLOSED
locked compound campaign	FAIL-CLOSED
portfolio target 12 / 17	NOT MET
ranking enabled	false

Card legend for pages 9–13. CC contract; SL source ledger; GEN generator; PROP molecule property; MDG molecule design gate; MCA model-card authority; PROV run provenance; BAL formulation arithmetic; CRO packet; IN evidence intake; RT red-team audit; RPT report packager. A scalar is shown only where the module emits one. "—" is an abstention, not zero. Every card has final score — and rank — because the system is deliberately non-ranking.

08 / FINAL OBSERVED CARDS

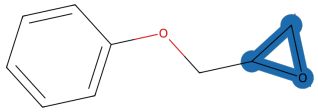
Epoxide-bearing proposal probes / 01–04

BLOCKED_PRECAMPAIGN

EA-P01 Phenyl glycidyl ether

AUDIT_ONLY / OPTIMIZER: NO

EPOXIDE RESIN / UNMADE AUDIT PROBE



SMILES `c1ccccc1OCC2CO2`

CC	SCORED 1.00	SL	FAIL 0.00	GEN	ABSTAIN —
PROP	ABSTAIN —	MDG	FAIL —	MCA	ABSTAIN —
PROV	SCORED —	BAL	ABSTAIN —	CRO	ABSTAIN —
IN	FAIL 0.00	RT	SCORED 1.00	RPT	ABSTAIN —

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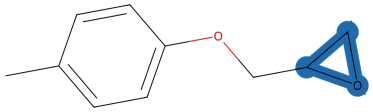
FINAL SCORE — / RANK — / DISPOSITION BLOCKED; EXACT CHEMISTRY + EVIDENCE REQUIRED

BLOCKED_PRECAMPAIGN

EA-P02 p-Cresyl glycidyl ether

AUDIT_ONLY / OPTIMIZER: NO

EPOXIDE RESIN / UNMADE AUDIT PROBE



SMILES `Cc1ccc(OCC2CO2)cc1`

CC	SCORED 1.00	SL	FAIL 0.00	GEN	ABSTAIN —
PROP	ABSTAIN —	MDG	FAIL —	MCA	ABSTAIN —
PROV	SCORED —	BAL	ABSTAIN —	CRO	ABSTAIN —
IN	FAIL 0.00	RT	SCORED 1.00	RPT	ABSTAIN —

RESULT SHA-256 `bfd3971653f8c60ab17b9ec0a2fbf244b4b0aefe68a652aef1fd254a5698b99e`

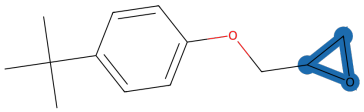
FINAL SCORE — / RANK — / DISPOSITION BLOCKED; EXACT CHEMISTRY + EVIDENCE REQUIRED

BLOCKED_PRECAMPAIGN

EA-P03 4-tert-Butylphenyl glycidyl ether

AUDIT_ONLY / OPTIMIZER: NO

EPOXIDE RESIN / UNMADE AUDIT PROBE



SMILES `CC(C)(C)c1ccc(OCC2CO2)cc1`

CC	SCORED 1.00	SL	FAIL 0.00	GEN	ABSTAIN —
PROP	ABSTAIN —	MDG	FAIL —	MCA	ABSTAIN —
PROV	SCORED —	BAL	ABSTAIN —	CRO	ABSTAIN —
IN	FAIL 0.00	RT	SCORED 1.00	RPT	ABSTAIN —

RESULT SHA-256 `9ad95d49d80dc46bb4c2664a4493b426078a20362d6d107e066f05848074d487`

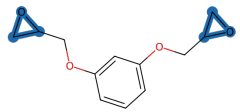
FINAL SCORE — / RANK — / DISPOSITION BLOCKED; EXACT CHEMISTRY + EVIDENCE REQUIRED

BLOCKED_PRECAMPAIGN

EA-P04 Resorcinol diglycidyl ether

AUDIT_ONLY / OPTIMIZER: NO

EPOXIDE RESIN / UNMADE AUDIT PROBE



SMILES `c1cc(OCC2CO2)cc(OCC3CO3)c1`

CC	SCORED 1.00	SL	FAIL 0.00	GEN	ABSTAIN —
PROP	ABSTAIN —	MDG	FAIL —	MCA	ABSTAIN —
PROV	SCORED —	BAL	ABSTAIN —	CRO	ABSTAIN —
IN	FAIL 0.00	RT	SCORED 1.00	RPT	ABSTAIN —

RESULT SHA-256 `8ad50eb1fb5707d262030937c5ae71db4056454c4a0e2c633e3180f1e2fbc7a1`

FINAL SCORE — / RANK — / DISPOSITION BLOCKED; EXACT CHEMISTRY + EVIDENCE REQUIRED

08 / FINAL OBSERVED CARDS

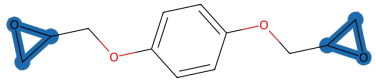
Epoxide-bearing proposal probes / 05–08

EA-P05 Hydroquinone diglycidyl ether

BLOCKED_PRECAMPAIGN

AUDIT_ONLY / OPTIMIZER: NO

EPOXIDE RESIN / UNMADE AUDIT PROBE



SMILES `c1cc(OCC2CO2)ccc1OCC3CO3`

CC	SCORED 1.00	SL	FAIL 0.00	GEN	ABSTAIN —
PROP	ABSTAIN —	MDG	FAIL —	MCA	ABSTAIN —
PROV	SCORED —	BAL	ABSTAIN —	CRO	ABSTAIN —
IN	FAIL 0.00	RT	SCORED 1.00	RPT	ABSTAIN —

RESULT SHA-256 `67643800a11dc423d6612345236220085c133b064d6006b0dab77ebc1fa3d782`

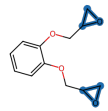
FINAL SCORE — / RANK — / DISPOSITION BLOCKED; EXACT CHEMISTRY + EVIDENCE REQUIRED

EA-P06 Catechol diglycidyl ether

BLOCKED_PRECAMPAIGN

AUDIT_ONLY / OPTIMIZER: NO

EPOXIDE RESIN / UNMADE AUDIT PROBE



SMILES `c1ccc(OCC2CO2)c(OCC3CO3)c1`

CC	SCORED 1.00	SL	FAIL 0.00	GEN	ABSTAIN —
PROP	ABSTAIN —	MDG	FAIL —	MCA	ABSTAIN —
PROV	SCORED —	BAL	ABSTAIN —	CRO	ABSTAIN —
IN	FAIL 0.00	RT	SCORED 1.00	RPT	ABSTAIN —

RESULT SHA-256 `4a82a2e78b6086c4a2e093ebf6dfd2ae163a131828f718de3675cdabaa8699eb`

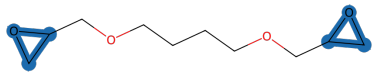
FINAL SCORE — / RANK — / DISPOSITION BLOCKED; EXACT CHEMISTRY + EVIDENCE REQUIRED

EA-P07 1,4-Butanediol diglycidyl ether

BLOCKED_PRECAMPAIGN

AUDIT_ONLY / OPTIMIZER: NO

EPOXIDE RESIN / UNMADE AUDIT PROBE



SMILES `C1OC1COCCCCOCC2CO2`

CC	SCORED 1.00	SL	FAIL 0.00	GEN	ABSTAIN —
PROP	ABSTAIN —	MDG	FAIL —	MCA	ABSTAIN —
PROV	SCORED —	BAL	ABSTAIN —	CRO	ABSTAIN —
IN	FAIL 0.00	RT	SCORED 1.00	RPT	ABSTAIN —

RESULT SHA-256 `0731f4a15d2128ad5eb4891135b0dc86110758f136e0a61dabcb2777fe425966`

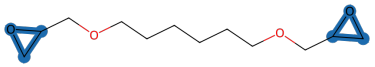
FINAL SCORE — / RANK — / DISPOSITION BLOCKED; EXACT CHEMISTRY + EVIDENCE REQUIRED

EA-P08 1,6-Hexanediol diglycidyl ether

BLOCKED_PRECAMPAIGN

AUDIT_ONLY / OPTIMIZER: NO

EPOXIDE RESIN / UNMADE AUDIT PROBE



SMILES `C1OC1COCCCCCCOCC2CO2`

CC	SCORED 1.00	SL	FAIL 0.00	GEN	ABSTAIN —
PROP	ABSTAIN —	MDG	FAIL —	MCA	ABSTAIN —
PROV	SCORED —	BAL	ABSTAIN —	CRO	ABSTAIN —
IN	FAIL 0.00	RT	SCORED 1.00	RPT	ABSTAIN —

RESULT SHA-256 `c5d42be58495ade0d056db2131014ee7ad0b976e1a983ba2369541854871e03e`

FINAL SCORE — / RANK — / DISPOSITION BLOCKED; EXACT CHEMISTRY + EVIDENCE REQUIRED

08 / FINAL OBSERVED CARDS

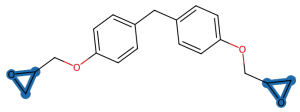
Epoxide models 09–10 / reactive amines 11–12

BLOCKED_PRECAMPAIGN

EA-P09 Bisphenol F diglycidyl ether model

AUDIT_ONLY / OPTIMIZER: NO

EPOXIDE RESIN / UNMADE AUDIT PROBE



SMILES `c1cc(OCC2CO2)ccc1Cc3ccc(OCC4CO4)cc3`

CC	SCORED 1.00	SL	FAIL 0.00	GEN	ABSTAIN —
PROP	ABSTAIN —	MDG	FAIL —	MCA	ABSTAIN —
PROV	SCORED —	BAL	ABSTAIN —	CRO	ABSTAIN —
IN	FAIL 0.00	RT	SCORED 1.00	RPT	ABSTAIN —

RESULT SHA-256 `c8247afff721486669f4c0e6a3a917cd9d34ba7dc426d48f7e3ca400dfdbc5f0`

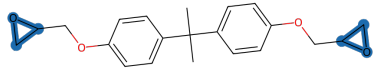
FINAL SCORE — / RANK — / DISPOSITION BLOCKED; EXACT CHEMISTRY + EVIDENCE REQUIRED

BLOCKED_PRECAMPAIGN

EA-P10 Bisphenol A diglycidyl ether model

AUDIT_ONLY / OPTIMIZER: NO

EPOXIDE RESIN / UNMADE AUDIT PROBE



SMILES `CC(c1ccc(OCC2CO2)cc1)(c3ccc(OCC4CO4)cc3)C`

CC	SCORED 1.00	SL	FAIL 0.00	GEN	ABSTAIN —
PROP	ABSTAIN —	MDG	FAIL —	MCA	ABSTAIN —
PROV	SCORED —	BAL	ABSTAIN —	CRO	ABSTAIN —
IN	FAIL 0.00	RT	SCORED 1.00	RPT	ABSTAIN —

RESULT SHA-256 `e9ee46ffcefd7ae8d389afc1dacc7c6ccb67c7767583ea0e9b37a7957057aced`

FINAL SCORE — / RANK — / DISPOSITION BLOCKED; EXACT CHEMISTRY + EVIDENCE REQUIRED

BLOCKED_PRECAMPAIGN

EA-P11 Ethylenediamine

AUDIT_ONLY / OPTIMIZER: NO

REACTIVE AMINE HARDENER / UNMADE AUDIT PROBE



SMILES `NCCN`

CC	SCORED 1.00	SL	FAIL 0.00	GEN	ABSTAIN —
PROP	ABSTAIN —	MDG	FAIL —	MCA	ABSTAIN —
PROV	SCORED —	BAL	ABSTAIN —	CRO	ABSTAIN —
IN	FAIL 0.00	RT	SCORED 1.00	RPT	ABSTAIN —

RESULT SHA-256 `c084c38073f79d553f89a561bd8001ae1bea8867b0e04a4efa6be1e7cae49d94`

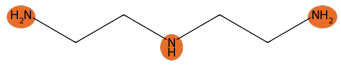
FINAL SCORE — / RANK — / DISPOSITION BLOCKED; EXACT CHEMISTRY + EVIDENCE REQUIRED

BLOCKED_PRECAMPAIGN

EA-P12 Diethylenetriamine

AUDIT_ONLY / OPTIMIZER: NO

REACTIVE AMINE HARDENER / UNMADE AUDIT PROBE



SMILES `NCCNCCN`

CC	SCORED 1.00	SL	FAIL 0.00	GEN	ABSTAIN —
PROP	ABSTAIN —	MDG	FAIL —	MCA	ABSTAIN —
PROV	SCORED —	BAL	ABSTAIN —	CRO	ABSTAIN —
IN	FAIL 0.00	RT	SCORED 1.00	RPT	ABSTAIN —

RESULT SHA-256 `7efdddb2763eb404b66e00259a78d61893e0891039ec4c652443ccb23cb2ff34`

FINAL SCORE — / RANK — / DISPOSITION BLOCKED; EXACT CHEMISTRY + EVIDENCE REQUIRED

08 / FINAL OBSERVED CARDS

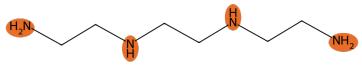
Reactive-amine proposal probes / 13–16

BLOCKED_PRECAMPAIGN

EA-P13 Triethylenetetramine

AUDIT_ONLY / OPTIMIZER: NO

REACTIVE AMINE HARDENER / UNMADE AUDIT PROBE



SMILES NCCNCCNCCN

CC	SCORED 1.00	SL	FAIL 0.00	GEN	ABSTAIN —
PROP	ABSTAIN —	MDG	FAIL —	MCA	ABSTAIN —
PROV	SCORED —	BAL	ABSTAIN —	CRO	ABSTAIN —
IN	FAIL 0.00	RT	SCORED 1.00	RPT	ABSTAIN —

RESULT SHA-256 0c3963117a6f2ef02b265a1c20a2c99fa23cbe2f14bd6691a7c6655de1b2e95f

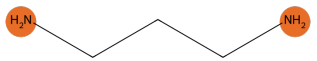
FINAL SCORE — / RANK — / DISPOSITION BLOCKED; EXACT CHEMISTRY + EVIDENCE REQUIRED

BLOCKED_PRECAMPAIGN

EA-P14 1,3-Diaminopropane

AUDIT_ONLY / OPTIMIZER: NO

REACTIVE AMINE HARDENER / UNMADE AUDIT PROBE



SMILES NCCCN

CC	SCORED 1.00	SL	FAIL 0.00	GEN	ABSTAIN —
PROP	ABSTAIN —	MDG	FAIL —	MCA	ABSTAIN —
PROV	SCORED —	BAL	ABSTAIN —	CRO	ABSTAIN —
IN	FAIL 0.00	RT	SCORED 1.00	RPT	ABSTAIN —

RESULT SHA-256 696ab3a8eee2837929d36a153e335974ac9a562b87047c16925d603d706f9510

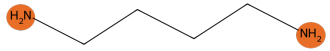
FINAL SCORE — / RANK — / DISPOSITION BLOCKED; EXACT CHEMISTRY + EVIDENCE REQUIRED

BLOCKED_PRECAMPAIGN

EA-P15 1,4-Diaminobutane

AUDIT_ONLY / OPTIMIZER: NO

REACTIVE AMINE HARDENER / UNMADE AUDIT PROBE



SMILES NCCCCN

CC	SCORED 1.00	SL	FAIL 0.00	GEN	ABSTAIN —
PROP	ABSTAIN —	MDG	FAIL —	MCA	ABSTAIN —
PROV	SCORED —	BAL	ABSTAIN —	CRO	ABSTAIN —
IN	FAIL 0.00	RT	SCORED 1.00	RPT	ABSTAIN —

RESULT SHA-256 153dfa385dc0ab46acc1841defe89c6b3d3d1c447b471e89eefc3619751d67d3

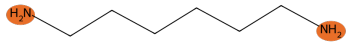
FINAL SCORE — / RANK — / DISPOSITION BLOCKED; EXACT CHEMISTRY + EVIDENCE REQUIRED

BLOCKED_PRECAMPAIGN

EA-P16 1,6-Diaminohexane

AUDIT_ONLY / OPTIMIZER: NO

REACTIVE AMINE HARDENER / UNMADE AUDIT PROBE



SMILES NCCCCCCN

CC	SCORED 1.00	SL	FAIL 0.00	GEN	ABSTAIN —
PROP	ABSTAIN —	MDG	FAIL —	MCA	ABSTAIN —
PROV	SCORED —	BAL	ABSTAIN —	CRO	ABSTAIN —
IN	FAIL 0.00	RT	SCORED 1.00	RPT	ABSTAIN —

RESULT SHA-256 dd9668d8720aff0f6d22537a8c8a44f9759b8e56f3c6bc38e94b27c0c414e707

FINAL SCORE — / RANK — / DISPOSITION BLOCKED; EXACT CHEMISTRY + EVIDENCE REQUIRED

08 / FINAL OBSERVED CARDS

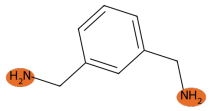
Reactive-amine proposal probes / 17–20

EA-P17 m-Xylylenediamine

BLOCKED_PRECAMPAIGN

AUDIT_ONLY / OPTIMIZER: NO

REACTIVE AMINE HARDENER / UNMADE AUDIT PROBE



SMILES NCc1cccc(CN)c1

CC	SCORED 1.00	SL	FAIL 0.00	GEN	ABSTAIN —
PROP	ABSTAIN —	MDG	FAIL —	MCA	ABSTAIN —
PROV	SCORED —	BAL	ABSTAIN —	CRO	ABSTAIN —
IN	FAIL 0.00	RT	SCORED 1.00	RPT	ABSTAIN —

RESULT SHA-256 69fa6fba010b893675fc7c00d59fdc767e3ab81ff0abd7259c619de4ca6064b8

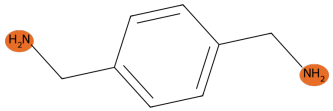
FINAL SCORE — / RANK — / DISPOSITION BLOCKED; EXACT CHEMISTRY + EVIDENCE REQUIRED

EA-P18 p-Xylylenediamine

BLOCKED_PRECAMPAIGN

AUDIT_ONLY / OPTIMIZER: NO

REACTIVE AMINE HARDENER / UNMADE AUDIT PROBE



SMILES NCc1ccc(CN)cc1

CC	SCORED 1.00	SL	FAIL 0.00	GEN	ABSTAIN —
PROP	ABSTAIN —	MDG	FAIL —	MCA	ABSTAIN —
PROV	SCORED —	BAL	ABSTAIN —	CRO	ABSTAIN —
IN	FAIL 0.00	RT	SCORED 1.00	RPT	ABSTAIN —

RESULT SHA-256 eca0cbb33c0a66d61f04312098cc237d00f393275d910bf0cc15c76d8a373584

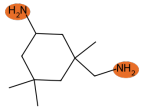
FINAL SCORE — / RANK — / DISPOSITION BLOCKED; EXACT CHEMISTRY + EVIDENCE REQUIRED

EA-P19 Isophorone diamine model

BLOCKED_PRECAMPAIGN

AUDIT_ONLY / OPTIMIZER: NO

REACTIVE AMINE HARDENER / UNMADE AUDIT PROBE



SMILES CC1(C)CC(N)CC(C)(CN)C1

CC	SCORED 1.00	SL	FAIL 0.00	GEN	ABSTAIN —
PROP	ABSTAIN —	MDG	FAIL —	MCA	ABSTAIN —
PROV	SCORED —	BAL	ABSTAIN —	CRO	ABSTAIN —
IN	FAIL 0.00	RT	SCORED 1.00	RPT	ABSTAIN —

RESULT SHA-256 4339aaff7cfd14c4983c929eece59b1a4c1a94d862d9b869186430e76c52bcc3

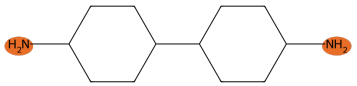
FINAL SCORE — / RANK — / DISPOSITION BLOCKED; EXACT CHEMISTRY + EVIDENCE REQUIRED

EA-P20 4,4'-Methylenebis(cyclohexylamine)

BLOCKED_PRECAMPAIGN

AUDIT_ONLY / OPTIMIZER: NO

REACTIVE AMINE HARDENER / UNMADE AUDIT PROBE



SMILES NC1CCC(CC1)C2CCC(CC2)N

CC	SCORED 1.00	SL	FAIL 0.00	GEN	ABSTAIN —
PROP	ABSTAIN —	MDG	FAIL —	MCA	ABSTAIN —
PROV	SCORED —	BAL	ABSTAIN —	CRO	ABSTAIN —
IN	FAIL 0.00	RT	SCORED 1.00	RPT	ABSTAIN —

RESULT SHA-256 8d8bd1d6931fcc22b4d7da42124bcd37be74c0c1f09bbb59c908391c7b5c3055

FINAL SCORE — / RANK — / DISPOSITION BLOCKED; EXACT CHEMISTRY + EVIDENCE REQUIRED

09 / DECISION RECORD

The right next milestone is evidence completion, not weaker gates

Release decision

Do not promote the four modules today. Keep all four `LOCKED_INTERNAL`, `AUDIT_ONLY`, non-ranking, non-procurement, non-safety, and non-release. Keep the portfolio at **8 Phase 0 / 21 pre-Phase-0**. No campaign result, rank, synthesis recommendation, or CRO execution authorization exists.

Critical-path action register

1. **Trust owner:** register three distinct authorized signer identities and verify possession/revocation state.
2. **Data owner:** acquire the exact 550-object ORD tree, write signed receipt, normalize, and publish exclusions.
3. **Build owner:** build RDKit C++ 2026.03.2 at the pinned commit; record compiler, dependencies, flags, source, and library hashes.
4. **Science owner:** execute the 300-structure property benchmark and leakage-safe grammar extraction.
5. **Campaign owner:** meet 24/8/8/6, then create 12 balances and 12 draft packets without threshold relaxation.
6. **Independent reviewer:** seal/reveal attacks, reimplement critical checks, replay clean runs, and sign review.
7. **Portfolio owner:** move exactly one target module at a time in the prescribed order, auditing globally after each.

What the 20 cards prove—and do not prove

They prove	Strict candidate contracts score 1.00; source/evidence deficiencies are visible; current red-team policy control scores 1.00; all controls remain optimizer-ineligible; three-lane composite traversal is deterministic; result hashes replay; current control-plane rate is measurable.
They do not	Prove ORD lineage, uniqueness, exact authoritative identity, formula, MW, EEW/AHEW, reactive counts, safety, availability, novelty, synthesis feasibility, coating performance, qualification, procurement suitability, or CRO execution.

Reproduction anchors

- Probe manifest: `reports/data/epoxy-amine-20-audit-probes.v1.json`
- Structure render receipt: `reports/assets/compound-structures/render-manifest.json`
- Card command: `compound audit-cards -manifest <probe-manifest>`
- Campaign preregistration: `config/program/epoxy-amine-compound-campaign.v1.json`
- v4 catalog: `feedback/v4/catalog.json`
- RDKit build recipe: `ops/rdkit/Dockerfile`

Bottom line

The implementation now makes the scientifically honest outcome easy: run when evidence is present, fail closed when it is not, preserve every reject, and never turn a diagnostic into authority. The next credible report should replace these blocked probes with at least 24 ORD-lineaged generated structures and 12 verified draft CRO packets—or report the campaign's negative result without promotion.

Prepared from repository state and local command evidence on 19 July 2026 UTC. No external signer, laboratory, procurement, safety, product-performance, or CRO-execution claim is made. Official inputs named by the preregistration: ORD data commit `ddb0d257...`; ord-schema 0.6.3 commit `0a6c925...`; RDKit release 2026.03.2 commit `42fe8c95...`; ChEBI release 252.